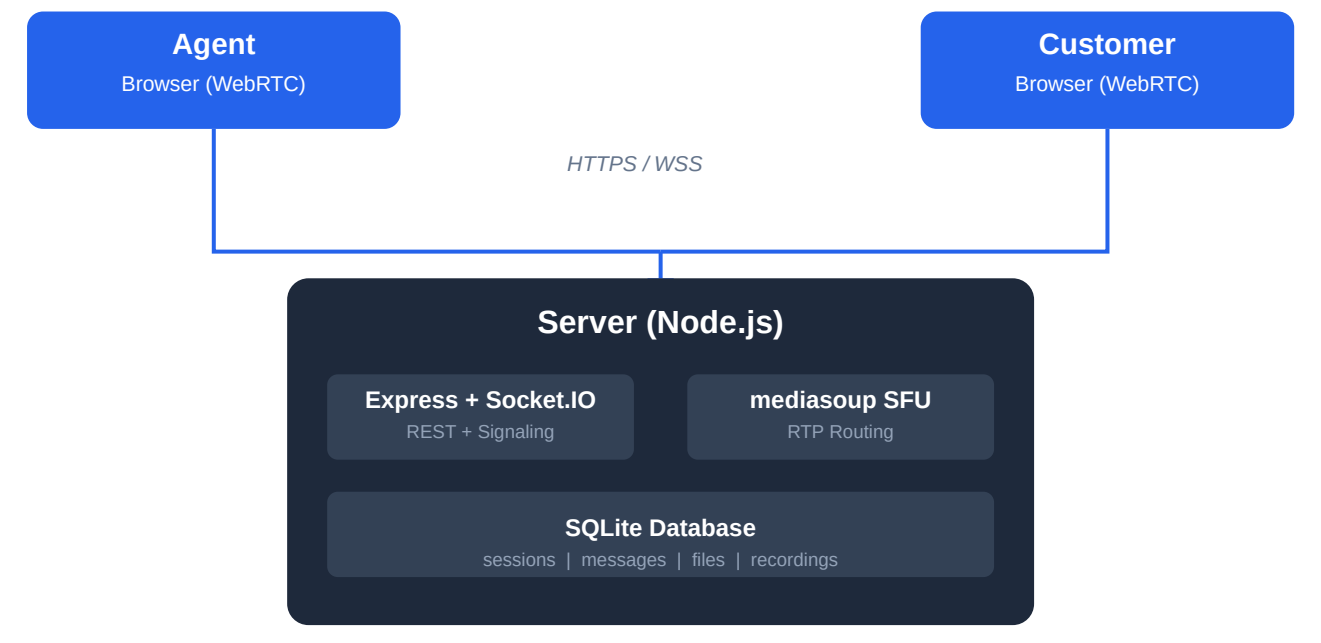


Architecture Overview

All audio and video flows through the server-side mediasoup SFU. Browsers never connect directly to each other.



Tech Stack

Media Server:	mediasoup v3 (SFU) — all RTP routed server-side, no P2P
Signaling:	Socket.IO — join, produce, consume, chat, session lifecycle
REST API:	Express.js — auth, sessions, files, admin, metrics
Database:	SQLite (better-sqlite3) — WAL mode, 6 tables, indexed
Authentication:	JWT + bcryptjs — role-based (agent / admin / customer)
Frontend:	React 18 + Vite + Tailwind CSS
WebRTC Client:	mediasoup-client v3 — send/recv WebRtcTransport
Recording:	MediaRecorder API (browser) !' WebM blob !' server upload
Metrics:	prom-client — Prometheus-compatible /metrics endpoint

WebRTC Signal Flow

- 1 join-room !' server validates JWT / invite token !' emits joined
- 2 get-rtp-capabilities !' server returns router codecs !' client loads Device
- 3 create-transport (send) !' server creates WebRtcTransport !' client connects
- 4 create-transport (recv) !' server creates WebRtcTransport !' client connects
- 5 produce !' client sends track !' server creates Producer !' broadcasts new-producer
- 6 consume !' peer requests Producer !' server creates Consumer (paused)
- 7 resume-consumer !' Consumer unpaused !' media flows

Login Credentials

Admin

Email **admin@demo.com**
Password **admin123**

Agent 1

Email **agent1@demo.com**
Password **agent123**

Agent 2

Email **agent2@demo.com**
Password **agent123**

Customer Access

Customer (no login required)

Customers join via invite link generated by the agent.

Flow: Agent creates session !' copies invite link !' customer opens link !' enters name !' joins call

Key Endpoints

POST	/api/auth/login	Login (email + password)
POST	/api/sessions	Create support session (agent)
GET	/api/sessions	List agent sessions
GET	/api/invite/:token	Validate customer invite token
POST	/api/files/upload	Upload file to session
GET	/api/admin/sessions	All sessions (admin)
DELETE	/api/admin/sessions/:id	Force-end session (admin)
GET	/metrics	Prometheus metrics
GET	/health	Server health check
GET	/debug/rooms	Live room state (dev)

Prometheus Metrics

atomquest_active_sessions_total	Gauge	Currently active sessions
atomquest_connected_participants	Gauge	Connected socket peers
atomquest_sessions_created_total	Counter	Total sessions ever created
atomquest_messages_sent_total	Counter	Chat messages sent
atomquest_errors_total	Counter	Server errors by type
atomquest_recordings_total	Counter	Completed recordings

